



IES E-CAMPUS: FACULTY OS & TEACHER MANAGEMENT MANUAL



Executive Presentation Deck & Faculty Cockpit Demonstration Guide

Document Purpose: Presentation Slide Deck for Evaluators, Faculty Mentors & Academic Management
Target Audience: Professors, HODs, Deans, Academic Directors & Accreditation Evaluators
Institution: IES College of Technology / IES University, Bhopal (RGPV & AICTE Approved)
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[Slide 1] Executive Overview: The Digital Faculty OS

1. The Real-College Problem

Faculty members in Indian technical universities face severe administrative burdens:

- Calling roll calls for 60 students per lecture wastes 15 minutes of every 50-minute teaching hour.
- Manually calculating monthly attendance percentages in thick paper register books.
- Signing paper registration forms, bonafide slips, and lab records by hand.
- Dealing with timetable clashes or absent colleagues without automated substitution tools.

2. The IES Faculty OS Solution

Faculty OS consolidates all administrative, teaching, and mentorship duties into a single mobile dashboard. Taking attendance takes under 20 seconds, timetable updates sync instantly, and students receive automated notifications without paperwork.

FACULTY MANAGEMENT OS		
TEACHING SUITE	ADMINISTRATIVE	BATCH ENGAGEMENT
• 1-Tap Attendance	• Student Approval	• Class Discussion Groups
• Live Broadcasts	• Sem Registrations	• Assignment Grading Desk
• Master Timetable	• Bonafide Sign-off	• Timed Quizzes Evaluation
• Notes Publishing	• Grievance Action	• Notice Broadcasts

Presenter Talking Points

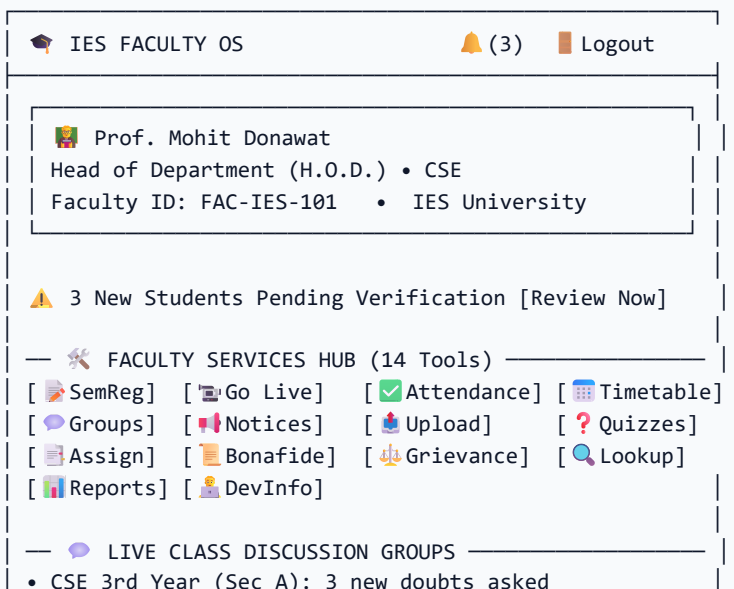
"Good morning respected evaluators. Teaching is an intellectual pursuit, but professors often spend a third of their time filling paper registers and signing forms. Faculty OS is engineered to eliminate clerical friction so professors can focus on teaching and mentorship."

 [SCREENSHOT PLACEHOLDER: Faculty OS Splash & Professor Onboarding Screen]

[Slide 2] Faculty Command Dashboard & 14-Tool Services Hub

1. Exact Screen Architecture

- Hero Faculty Identity Card:**
- Displays Professor Name, Designation (*Head of Department / Assistant Professor*), Department (*CSE*), and Faculty ID (*FAC-IES-101*).
- Pending Students Alert Banner:**
- Glowing amber warning card displaying unverified student applications with direct review shortcut.
- 14-Tool Faculty Services Grid:**
- Quick 1-tap navigation for:
 -  **Sem Register** (Approve enrollment forms)
 -  **Go Live** (Broadcast live lecture)
 -  **Attendance** (Mark lecture roll call)
 -  **Timetable** (Manage class routine)
 -  **Class Groups** (Batch discussion hub)
 -  **Notices** (Broadcast department circulars)
 -  **Upload Notes** (Add PDFs/PPTs to library)
 -  **Quizzes** (Create online tests)
 -  **Assignments** (Grade submissions)
 -  **Bonafide** (Attest certificates)
 -  **Grievance** (Resolve student tickets)
 -  **Lookup** (Search individual student file)
 -  **Reports** (Attendance analytics)
 -  **Developer** (Architecture specs)
- Live Class Discussion Groups Box:**
- Live stream of batch doubt channels showing recent messages and quick-reply action buttons.



Presenter Talking Points

"Everything a professor needs is accessible in one tap. The dashboard highlights pending approvals, active class doubts, and daily tools in an uncluttered, high-contrast interface."

 [SCREENSHOT PLACEHOLDER: Faculty Home Dashboard with 14-Tool Grid & Pending Alert]

[Slide 3] Student Account Verification & Identity Approval Gate

1. Step-by-Step Working Mechanism

- **Trigger:** New student registers on the app.
- **Faculty Alert:** The alert banner shows `X Students Pending Approval`.
- **Review Workflow:**
 - Professor taps "Review".
 - Inspects applicant credentials: Full Name, University Roll Number, RGPV Enrollment, Branch, Year, and Section.
- **Decision Actions:**
 -  **Approve (One-Tap):** Updates database to `isApproved: true`. The student's dashboard unlocks immediately, and they are enrolled in the class attendance register and batch discussion group.
 -  **Reject:** Deletes bogus accounts, preventing outsiders from accessing college resources.

Presenter Talking Points

"This gatekeeper mechanism prevents unauthorized users from accessing campus streams or student directories. Every student must be authenticated by their department faculty before their account is activated."

 [SCREENSHOT PLACEHOLDER: Pending Student Approvals Screen with Verify and Reject Buttons]

[Slide 4] One-Tap Classroom Attendance Register (20-Second Roll Call)

1. Exact Step-by-Step Operation

1. **Context Selection (5 seconds):**
2. Professor selects Branch (`CSE`), Year (`3rd Year`), Semester (`5th Sem`), Section (`A`), Subject (`Operating Systems`), and Period (`Period 2`).
3. **Auto-Load Roster:**
4. The enrolled class roster loads in alphabetical order by Roll Number.
5.  **The One-Tap Present Action (2 seconds):**
6. Professor taps "**Mark All Present**". All 60 student cards turn green (`PRESENT`) instantly.
7. **Marking Absentees (10 seconds):**
8. Professor scans the classroom and taps only the 2 or 3 absent students, switching their cards to red (`ABSENT`).
9. **Submit & Instant Push (3 seconds):**
10. Professor taps "**Submit Attendance**".
11. Record writes to Firestore collection `attendance` with subject, period, timestamp, and student attendance map.
12. **Total Duration: Under 20 seconds.**
13. Student percentages update in real time on their personal meters.

MARK ATTENDANCE: CSE 3rd Year • Sec A • Period 2

Subject: Operating Systems (CS-501)Date: Today

[⚡ Mark All Present (60 Students)]

Roll 220101 - Amit Kumar

PRESENT

Absent

Roll 220102 - Anjali Verma

PRESENT

Absent

Roll 220103 - Deepak Sharma

Present

ABSENT

Roll 220104 - Priya Patel

PRESENT

Absent

Summary: 58 Present, 2 Absent (96.7% Class Turnout)

[📄 Submit Attendance & Notify Absentees]

 Presenter Talking Points

"Calling 60 names wastes 15 minutes of every lecture. With our 'Mark All Present' mechanism, professors turn the entire class green in one tap, deselect only the absentees, and finish attendance in under 20 seconds."

 **[SCREENSHOT PLACEHOLDER: Attendance Marking Interface with Quick Toggle Buttons]**

[Slide 5] Attendance Analytics, Student Lookup & Detention Radar

1. Step-by-Step Working Mechanism

- **Instant Student Lookup:**
 - Search any student by Roll Number or Name.
 - Displays: Total lectures held vs attended, subject-wise percentage breakdown, and exact absentee dates.
- **Detention Radar:**
 - Filter across branch and semester to view all students below the mandatory 75% threshold.
 - Direct communication actions:
 -  **1-Tap Direct Call:** Dial student or parent phone number directly.
 -  **1-Tap Gmail Alert:** Send an attendance warning email via Gmail.
-  **Export RGPV Detention List:** 1-click export formatted for submission to the university exam cell.

 Presenter Talking Points

"HODs no longer need to spend days compiling defaulter lists at the end of the term. The detention radar shows at-risk students in real time and provides one-tap calling to notify parents before it's too late."

 **[SCREENSHOT PLACEHOLDER: Attendance Defaulter Report with Parent Call and Email Buttons]**

[Slide 6] Master Timetable Manager & Period Slot Scheduler

1. Step-by-Step Working Mechanism

- **Class Selector:** Choose Branch, Year, Semester, and Section.
- **Dual Operating Modes:**
 -  **Weekly Master (Daily Default):** Set standard recurring schedule from Monday to Saturday.
 -  **Today's Change / Substitute:** Modify day-specific lectures when a colleague is on leave or an extra lab is scheduled.
-  **1-Click "Auto-Fill Standard Curriculum":**
 - Automatically loads core semester subjects (OS, DBMS, CN, SE, Labs) based on AICTE/RGPV curricula across

all 6 days.

- **Interactive Period Editor:**
- Configure Period Number (1 to 8), Time Range, Subject, Faculty In-Charge, and Classroom/Lab Room number.
-  Edit and  Delete buttons on every card.
- **Notice Board Photo Attachment:** Upload a photo of the printed notice board schedule alongside structured period slots.

TIMETABLE MANAGER: CSE • 3rd Year • 5th Sem • Sec A

Mode: [ Weekly Master] [ Today's Substitute]

[ Auto-Fill Standard RGPV Curriculum Schedule]

[ Mon(6)] [Tue(6)] [Wed(6)] [Thu(6)] [Fri(6)]

Period 1 (09:00 - 10:00 AM)
Subject: Operating Systems (CS-501)
Room: Room 304 • Faculty: Prof. Mohit Donawat
[ Edit] [ Delete]

[ Add New Period Slot] [ Save & Publish]

Presenter Talking Points

"Timetables are living schedules. If a professor is absent today, a substitute teacher can be scheduled in 10 seconds. Students immediately see the substitute badge on their phones, eliminating empty classroom confusion."

 [SCREENSHOT PLACEHOLDER: Timetable Management Screen with Period Slot Cards and Mode Switcher]

[Slide 7] Semester Registration Approvals Portal (HOD Attestation Desk)

1. Step-by-Step Working Mechanism

- **Queue Management:** Filter by department and semester to view pending submissions.
- **Student Profile Card:**
 - Full Name & RGPV Enrollment Number
 - Previous SGPA & Cumulative CGPA
 - **Backlog / ATKT Indicator:** Highlights uncleared backlogs in red.
 - Co-curricular achievements & elective course choices.
 - Fee Payment Reference Number and verification status.
- **HOD Actions:**
 -  **Inspect PDF:** Preview the generated A4 five-part document.
 -  **Approve & Sign:** Stamps official digital HOD signature and crimson red seal.
 -  **Reject:** Returns form with specific feedback (e.g., "Fee receipt number unverified").

SEMESTER REGISTRATION APPROVALS (HOD DESK)

Filter: [Computer Science & Engg ▼] [5th Semester ▼]

 Rahul Sharma (0103CS221045)
SGPA: 8.42 • CGPA: 8.10 • Backlogs: NONE (0)
Achievements: Smart India Hackathon Winner
Elective: Advance Database Systems
Fee Status:  PAID (Ref: IES/FEE/2024/99128)
[ View Form PDF] [ Reject] [ Approve]

Presenter Talking Points

"HODs no longer need to manually sign hundreds of paper registration forms. The portal displays academic progression, backlogs, and fee payment status at a glance, allowing forms to be verified and attested in one tap."

 [SCREENSHOT PLACEHOLDER: HOD Semester Registration Queue with Student Inspection Cards]

[Slide 8] Live Lecture Broadcasting & Real-Time Classroom Studio

1. Step-by-Step Working Mechanism

- **Schedule:** Enter Subject, Topic, Date, and Time; select target batch.
- **Go Live Action:**
 - Mounts live camera stream (or embedded YouTube Live / WebRTC broadcast).
 - Status updates to **LIVE** in database, triggering alert banners on student dashboards.
- **Live Doubts Stream:**
 - Student questions display in a dedicated panel during the lecture.
 - Professor answers doubts in real time; responses carry golden  **FACULTY** badges.
- **Post-Lecture Materials:** Upload slide presentations (PPT/PDF) directly beneath the video player.

Presenter Talking Points

"Whether for revision sessions or campus holidays, professors can broadcast live lectures directly to their target batches. Live doubt tracking and post-lecture handout distribution are integrated directly into the stream."

 [SCREENSHOT PLACEHOLDER: Teacher Go Live Broadcasting Studio with Student Doubts Panel]

[Slide 9] Study Material & Digital Library Publishing

1. Step-by-Step Working Mechanism

- **Upload Details:** Enter Title, Unit/Topic, and select target Department, Year, and Subject.
- **File Attachment:** Select PDF, Word doc, or PowerPoint file directly from phone storage.
- **Instant Cloud Distribution:**
 - File is processed as a Base64 Data URI and saved directly into Firestore.
 - Appears immediately in the digital library of every enrolled student in that batch.
 - Tracks student view and download counts.

Presenter Talking Points

"Publishing lecture notes takes 10 seconds. Study materials are automatically categorized by subject and semester on the student side, ending reliance on photocopies."

 [SCREENSHOT PLACEHOLDER: Teacher Library Upload Form with Branch/Subject Selector]

[Slide 10] Quiz Creation & Automated Test Evaluation Engine

1. Step-by-Step Working Mechanism

- **Question Formulation:**
 - Input Question prompt.
 - Provide 4 Multiple Choice options and mark the correct answer radio button.
 - Add optional explanation for students to read after submission.
- **Timer & Target Controls:**
 - Set test duration (e.g., 20 minutes) and expiration deadline.
 - Select target batch (e.g. *CSE 3rd Year*).
- **Automated Grading:**
 - System evaluates all student attempts automatically.
 - Generates full class score sheets, average marks, and top performers with zero manual grading.

Presenter Talking Points

"Automated quizzes provide continuous evaluation without grading fatigue. Scores, detailed explanations, and class leaderboards are generated the moment the test expires."

 [SCREENSHOT PLACEHOLDER: Teacher Quiz Builder Screen with Multiple Choice Formulation]

[Slide 11] Assignment Publishing & Grading Desk (6-Month Archive)

1. Step-by-Step Working Mechanism

- **Dispatch:** Post assignment title, problem statement, max marks, deadline, and reference sheet.
- **Submission Tracking:** Live counter displays: `"42 / 60 Students Submitted"`.
- **Review & Grading Desk:**
 - Open any student's submitted scan or typed PDF.
 - Enter Marks (e.g., `18 / 20`) and type professor feedback ("*Good approach on question 3, improve time complexity*").
 - Grades sync to the student's personal review archive immediately.

Presenter Talking Points

"Managing assignments on paper creates storage chaos. The digital grading desk provides submission timestamp verification and keeps an auditable 6-month grading archive."

 [SCREENSHOT PLACEHOLDER: Assignment Grading Desk with Student Submission List]

[Slide 12] Bonafide Certificate Attestation & Dispatch Desk

1. Step-by-Step Working Mechanism

- **Receive Request:** Student applies for Bonafide with selected purpose (Scholarship, Loan, Bus Pass, Passport).
- **Verification:**
 - Faculty reviews candidate's attendance record and active semester enrollment.
- **Attest & Dispatch:**
 - One tap stamps the digital college header, attested photo, circular red seal, and authorized signature.
 - High-res A4 PDF is generated with an embedded verification QR code and delivered directly to the student portal.

Presenter Talking Points

"Bonafide requests that once took days of clerk routing are verified and dispatched in seconds, complete with institutional seal and QR code verification."

 [SCREENSHOT PLACEHOLDER: Teacher Bonafide Verification Desk with 1-Tap Sign & Dispatch]

[Slide 13] Class Discussion Groups & Mentorship Supervision

1. Step-by-Step Working Mechanism

- **Batch Group Creation:** Create targeted rooms (e.g., "CSE 3rd Year - OS Doubts") with quick title chips ("  OS Discussion", "  DBMS Queries").
- **Faculty Moderation:**
- Faculty messages carry golden  FACULTY cards, providing clear authority.
- **Anti-Tamper Protections:**
- Message deletion is permanently blocked.
- Edits are restricted to 60 seconds with an  tag, ensuring academic decorum.

Presenter Talking Points

"Academic discussions require decorum. Our discussion hub prevents message deletion, limits edits to 60 seconds, and highlights verified faculty responses."

 [SCREENSHOT PLACEHOLDER: Faculty Managing Class Discussion Group with Quick Topic Chips]

[Slide 14] Institutional Notice & Urgent Circular Broadcasting

1. Step-by-Step Working Mechanism

- **Priority Selector:**
-  **URGENT:** Triggers push notifications and high-priority red alert banners (e.g. heavy rain holiday, exam date change).
-  **NORMAL:** General academic circulars and event notifications.
- **Target Selection:** Broadcast to a single branch, a specific semester, or the entire college.
- **Document Attachment:** Attach official university circular orders (PDF/Images) directly.

Presenter Talking Points

"Critical circulars are broadcast directly to student phones in seconds, replacing physical paper notices that are easily missed."

 [SCREENSHOT PLACEHOLDER: Notice Creation Screen with Urgency Selector and Document Attachment]